



Opinion piece



Cite this article: Santander CG, Campelo dos Santos AL, Arnab SP, Fumagalli M, DeGiorgio M. 2026 Negative frequency-dependent selection: a positive outlook with deep learning. *Phil. Trans. R. Soc. B* **381**: 20250170.
<https://doi.org/10.1098/rstb.2025.0170>

Received: 6 June 2025

Accepted: 7 October 2025

One contribution of 18 to a theme issue ‘Exploring negative frequency dependent selection across levels: from genetics to ecology and back again’.

Subject Areas:

computational biology, evolution, genomics, bioinformatics

Keywords:

transfer learning, feature extraction, balancing selection

Authors for correspondence:

Matteo Fumagalli

e-mail: m.fumagalli@qmul.ac.uk

Michael DeGiorgio

e-mail: mdegior@fau.edu

[†]These authors contributed equally to the study.

Supplementary material is available online at
<https://doi.org/10.6084/m9.figshare.c.8447772>.

Negative frequency-dependent selection: a positive outlook with deep learning

Cindy Gilda Santander^{1,2,†}, Andre Luiz Campelo dos Santos^{1,†}, Sandipan Paul Arnab^{1,2}, Matteo Fumagalli^{4,5} and Michael DeGiorgio^{1,2,3}

¹Department of Electrical Engineering and Computer Science, ²Center for Omics Technologies and Data Engineering, and ³Department of Biomedical Engineering, Florida Atlantic University, Boca Raton, FL 33431, USA

⁴School of Biological and Behavioural Sciences, Queen Mary University of London, London E1 4NS, UK

⁵The Alan Turing Institute, London NW1 2DB, UK

CGS, 0000-0003-3021-6809; **ALCdS**, 0000-0002-5577-1967; **SPA**, 0000-0003-0827-5327; **MF**, 0000-0002-4084-2953; **MDG**, 0000-0003-4908-7234

Balancing selection is a mode of natural selection that maintains genetic diversity through various mechanisms, including negative frequency-dependent selection (NFDS). However, distinguishing the genomic signature of NFDS from those of other balancing selection modes, such as overdominance, remains a significant challenge. In this perspective, we outline strategies to improve the modelling of genomic patterns expected under NFDS, with the goal of better differentiating them from signals of neutrality and alternative selection processes. We demonstrate how resource-efficient deep transfer learning, combined with novel data preprocessing and the modelling of genomic autocovariation, can effectively detect and characterize NFDS using either phased or unphased genotypes, and with or without temporal data from ancient DNA. Finally, we offer practical recommendations for both empiricists and method developers on advancing the detection of NFDS in genomic data.

This article is part of the theme issue ‘Exploring negative frequency dependent selection across levels: from genetics to ecology and back again’.

1. Introduction

A central goal of evolutionary biology is to understand the myriad factors that have shaped the present-day distribution of variation among organisms. Key processes include the genetic effects of mutation and recombination, demographic history, ecological and environmental influences and the roles of adaptive traits. Dissecting these components often involves identifying genomic footprints characteristic of the underlying processes [1]. However, these genomic signals are often subtle, and advanced approaches are necessary to disentangle the conflicting signals of the many forces at work. Balancing selection [2], which adaptively maintains polymorphism in populations, is particularly difficult to decipher [3].

Early methods for uncovering adaptive events from genomic variation relied on summary statistics, which captured specific measurements of genome variation patterns [4–15]. These summaries are typically quick to compute and often grounded in theoretical models or based on trends observed from visualizing mean patterns in simulated data. However, mean signals alone often lack the resolution to effectively distinguish between adaptive and non-adaptive factors, largely because they do not capture the linear dependence (autocovariation) of genomic variation along the genome expected owing to linkage disequilibrium [16]. This is the case with frequency-based methods, whereas haplotype-based methods do capture proxies of linkage disequilibrium. However, they are often unable

to disentangle signals owing to demographic factors such as population fluctuations from non-demographic factors like natural selection [17–19].

Over the past two decades, the field has experienced a significant shift towards directly modelling genomic autocovariation when identifying adaptive footprints, primarily through maximum likelihood approaches [20–28]. While full likelihood models are generally computationally intractable, researchers have developed composite (or pseudo) likelihood frameworks that simplify the problem by assuming loci are independent. Nevertheless, these models explicitly account for the autocovariation between the putatively selected (target) site and each neighbouring locus, with the independence between the test site and other loci increasing with their physical distance.

These methods have been shown to achieve substantially higher power to detect adaptive events compared with summary statistic approaches, provided that their modelling assumptions are met. However, they can also be computationally intensive, as they require the optimization of multiple model parameters at each test location, which may span thousands of sites [20–30].

A key hurdle for both summary statistic and likelihood-based approaches in detecting balancing selection is that they are primarily tailored to long-term balancing selection [4–7,10,31–37], occurring for over more than $4N_e$ generations of selection [38], where N_e is the diploid effective population size. Long-term balancing selection is typically characterized by an increased density of within-species polymorphism relative to between-species substitution, and it produces a narrow genomic footprint owing to recombination eroding linkage disequilibrium over many generations. In some cases, this process may produce inter-species single nucleotide polymorphisms (SNPs), which can be challenging to identify without large samples from multiple species [39].

By contrast, the patterns associated with recent or transient balancing selection, occurring less than $4N_e$ generations ago [38,40], are less well understood. Moreover, it remains unclear how readily different modes of balancing selection, such as overdominance (OD) and negative frequency-dependent selection (NFDS), can be distinguished using measurements of contemporary genomic variation alone [41,42]. As previous work has demonstrated, OD can generate NFDS given that common alleles will more often present themselves as homozygotes than rare alleles will [41,43,44], making it more difficult for available methods to distinguish these modes of balancing selection when overlap exists. In the absence of a strong theoretical foundation, developing methods to detect recent or different forms of balancing selection is challenging, raising the question of whether additional types of data beyond modern sampled genomic variation are necessary.

The renaissance of artificial intelligence and machine learning over the past decade has resulted in a paradigm shift in characterizing adaptive and other evolutionary events [38–49]. Notably, although it can be difficult to write explicit summaries or derive models for certain evolutionary phenomena, it is relatively easy to simulate data from arbitrarily complex processes using modern software packages [50,51]. Hence, we can leverage simulated (or synthetic) genetic variation under different evolutionary scenarios to train machine learning models to discriminate among different events [47,48]. Simulation-based machine learning models have proven to be highly powerful and flexible in population genomics [52,53]. Among machine learning frameworks, deep learning is particularly effective at capturing spatial patterns [54], object detection such as recognizing edges [55] and generalizing across varying scales [56]. Moreover, deep learning can automatically extract these abstract features [55]. Its nonlinear structure allows it to serve as a universal approximator when provided with sufficient model parameters [57,58], and it can integrate data from heterogeneous sources [59].

However, a key challenge with deep learning approaches is that they are data-hungry, often requiring thousands or even millions of training examples to achieve adequate performance. Although genomic variation is relatively easy to simulate, generating enough replicates can still be time- and computation-intensive, especially for complex processes evolving over long genomic regions or those leaving cryptic signals, such as recent balancing selection [60,61]. This computational burden may limit the accessibility of such approaches for researchers operating with minimal resources and comes at a significant environmental cost in terms of carbon footprint [62]. For all these reasons, we still lack powerful and efficient methods to quantify the extent to which recent and transient balancing selection shapes genomic variation.

To date, few studies have used machine learning to detect balancing selection [41,46,48,49] and none have focused primarily on distinguishing NFDS from other types of balancing selection. In this study, we evaluate the potential of deep learning for detecting recent balancing selection using both modern and ancient DNA, as well as for distinguishing between OD and NFDS. To address the substantial resource requirements of deep learning approaches, we explore the machine learning technique known as transfer learning, which enables the extraction of useful features from a model pre-trained on a large, unrelated dataset and the use of these features to train downstream models with minimal additional data [63]. Transfer learning is implemented here via a tool that extracts transformed feature sets from natural images [64]. The learnt representations of a pre-trained model are then leveraged to extract features that are later used as input to a classifier to detect adaptive events, and a regressor to predict the underlying evolutionary parameters of such events. Here, transfer learning enhances the detection and classification of balancing selection patterns by equipping the classification model with rich features extracted from data that simulate NFDS and OD as input. Likewise, the extracted feature sets also allow the regressor to predict the equilibrium frequency at balanced polymorphisms that underlie both NFDS and OD. We test the performance of *TrIdent* [64], a tool that employs transfer learning as a feature extractor, in classifying different types of balancing selection in addition to predicting equilibrium frequencies at balanced polymorphisms, in the presence and absence of phased data, and when simultaneously analysing temporal data from ancient genomes. Based on our findings, we provide guidance for empiricists and methodologists on how to advance efforts to better characterize recent balancing selection, with a particular focus on NFDS.

2. Results

(a) Discriminating among evolutionary forces with modern haplotype variation

To explore the efficacy of transfer learning in disentangling the signatures left by OD, NFDS and neutrality, we initially trained the deep transfer learning-based classifier *TrIdent* [64] on simulated modern genomic samples (figure 1; see §4). *TrIdent* was originally designed for the binary problem of distinguishing between selective sweeps from positive selection and neutrality. A visual inspection of mean image representations of the haplotype alignments used to train *TrIdent* [64] suggests a clear pattern in the central windows that helps distinguish balancing selection from neutrality (electronic supplementary material, figure S1). This centre vertical strip pattern appears to be driven by the higher amount of intermediate frequency alleles around the target locus, which is to be expected in OD and NFDS. Lower pixel values (dark shades) indicate haplotypes (rows) in genomic windows (columns) with greater numbers of major alleles, and tall black regions suggest a high-frequency haplotype of major alleles. However, differences in mean patterns between OD and NFDS are more subtle, suggesting these forces may be inherently harder to differentiate. This difficulty is also reflected in the substantial overlap in their summary statistic distributions (electronic supplementary material, figure S2).

After pixel-wise standardization (i.e. transforming each pixel to have mean zero and a standard deviation of one across training images), differences among the three classes become clearer, with an almost opposite pattern for neutrality and NFDS for which negative pixel values in one class are positive in the other class (electronic supplementary material, figure S3). However, OD falls intermediate between the standardized variation observed from the other two classes (electronic supplementary material, figure S3), suggesting that OD will probably be the most difficult to distinguish among the three classes as it visually appears as a mixture of the other two classes. These mean qualitative patterns suggest that while classification is feasible, differentiating between the modes of balancing selection remains particularly challenging—with NFDS showing the greatest contrast to neutrality.

Consistent with these observations, our binary classifier effectively distinguishes neutrality from NFDS, achieving a high accuracy of 98.60% (electronic supplementary material, figure S4A). Classification accuracy for OD versus neutrality is lower at 85.05%, with a conservative bias towards labelling cases as neutral (electronic supplementary material, figure S4B). Distinguishing between OD and NFDS proves more difficult, with a reduced accuracy of 70.05% (electronic supplementary material, figure S4C). In this comparison, NFDS is more reliably detected (77.1% accuracy) than OD (63.0%), probably owing to the higher haplotype variation maintained in heterozygotes under OD. Moreover, NFDS patterns exhibit broader minor allele count variation within haplotype windows, whereas OD minor allele count variation falls within the NFDS minor allele count distribution. The NFDS minor allele count distribution results in NFDS input images spanning a wider intensity range (electronic supplementary material, figure S3), distinguishing NFDS from OD but rendering OD patterns less distinguishable. These trends are also reflected in the comparative power to discriminate between classes in each scenario (electronic supplementary material, figure S4D). Note that *TrIdent* performs a conservative step of filtering out singletons and doubletons from the genomic input data to ensure that no sequencing or mapping errors are included [64], thus removing potentially relevant information that could further help discriminate among the classes.

While informative, a trio of binary classifiers is not ideal for interpreting the evolutionary process underlying observed variation in a given genomic region. To address this barrier, we implemented a two-stage classification model based on this classifier trio: the first stage distinguishes neutrality from balancing selection, and the second differentiates between balancing selection modes if neutrality is not supported (see *Supplementary Methods* of the electronic supplementary material). This approach yields an accuracy of 63.39%, with class-specific detection rates of 95.50% for neutrality, 54.45% for NFDS and 40.23% for OD—mirroring the expected performance rankings (electronic supplementary material, figure S5).

We also explored whether a multiclass classifier trained to distinguish all three categories simultaneously would yield better results. This approach improved accuracy to 70.17%, primarily owing to a substantial gain in the detection of NFDS (93.90%), with only a modest decrease in the accuracy for OD (40.2%) but at the cost of detecting neutrality (76.4%) accurately (figure 2A). These outcomes align with the relative power to detect each process (electronic supplementary material, figure S6A) and the ability to distinguish each mode of balancing selection from neutrality (electronic supplementary material, figure S7A). We saw improved accuracy using a multi-class approach over a two-stage binary one and specifically recommend the multi-class model for disentangling modes of balancing selection and neutrality. Collectively, these findings support the promise of using transfer learning to detect NFDS from modern genomic data, while highlighting opportunities to improve performance, potentially through the integration of additional data sources.

(b) Refining performance with ancient genome variation

Detecting and distinguishing modes of balancing selection is challenging, but our results suggest that deep transfer learning represents a promising path forward. However, our findings thus far have relied only on a snapshot of genomic variation from present-day sampled individuals. Recent work has shown that incorporating temporal information into deep learning frameworks can enhance the detection of adaptive events [65]. We therefore trained classifiers in two additional settings: one using only ancient genomic variation sampled across eight time periods, and another combining modern genomic variation with this ancient data (figure 1; see §4).

When using only ancient samples and omitting modern data, overall accuracy declines slightly from 70.17% to 67.53% (compare figure 2A,B). Notably, this setting results in a substantial drop in the detection of NFDS—from 93.9 to 76.6%—which is offset by a significant improvement in the neutral detection rate, increasing from 76.4 to 89.3%. The highest classification

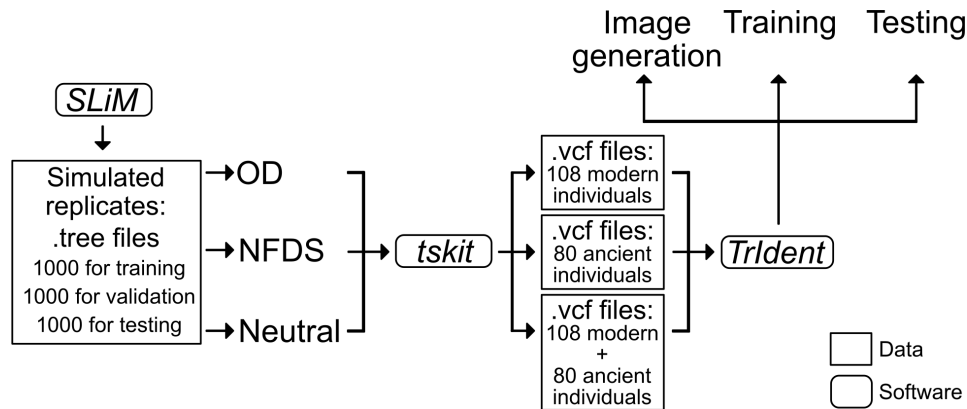


Figure 1. Schematic of our experimental pipeline, illustrating modern and ancient dataset generation for each of neutrality, OD and NFDS, followed by training and evaluation with *TrlDent* [64]. This pipeline resulted in models trained on three datasets: 108 modern diploid individuals; 80 ancient diploid individuals evenly sampled across eight epochs; a combination of these 108 modern and 80 ancient individuals.

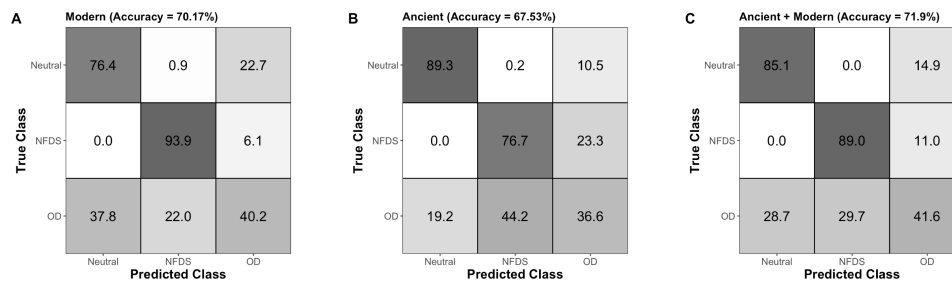


Figure 2. Classification rates and associated accuracies depicted by confusion matrices for distinguishing among evolutionary forces. Models trained and tested with (A) only modern genome variation; (B) only ancient genome variation; and (C) both modern and ancient genome variation.

accuracy of 71.9%, is achieved by including modern samples back in, which is a slight improvement on using modern samples alone (figure 2C). This improved accuracy is primarily owing to better detection of OD—41.6% compared with 40.2 and 36.6% in the other two settings—alongside intermediate detection rates for neutrality (85.1%) and NFDS (89%). Thus, combining ancient and modern variation produces a more balanced classifier. As expected, the power to distinguish between classes (electronic supplementary material, figure S6) and to detect balancing selection (electronic supplementary material, figure S7) aligns with the observed classification rates. While additional repetitions with different train/validation/test splits would be required to rigorously assess the generalizability of these findings, we highlight that adding ancient samples can alter the balance of error profiles.

(c) Estimating underlying adaptive parameters

Although the classification tasks of detecting balancing selection and distinguishing among its modes are important for characterizing the adaptive landscape of the genome, they overlook the estimation of adaptive parameter values—key elements for interpreting how adaptation has proceeded. Specifically, for a region confidently classified as undergoing balancing selection (regardless of mode), knowledge of the equilibrium frequency x^* is essential for forming a comprehensive picture of past evolutionary dynamics. To address this complementary inference task, we reframed *TrlDent* as a regression model to predict x^* . Notably, the estimate of x^* can be transformed into an estimate of the dominance parameter h when the classified mode is OD (see §4).

For combined modern and ancient variation, within the parameter ranges used for training and testing, the true parameter distributions are relatively uniform, as designed. Although the predicted distributions for x^* are wider than the true distribution (figure 3), they are more symmetric around their medians, with the predicted and true medians closely aligned. Moreover, the root mean squared error for predictions remains low—between 0.048 and 0.079 for predicting x^* . Predicted x^* values broadly track the true values for both NFDS and OD (electronic supplementary material, figure S8), which is further supported by the root mean squared error (RMSE) values obtained for NFDS and OD (figure 3). These results are mirrored when using modern variation alone, though with greater uncertainty in predictions compared with when information from ancient samples is jointly used (electronic supplementary material, figure S9). Thus, once a genomic region is identified as evolving under balancing selection, this regression approach within a deep transfer learning framework provides reliable estimates on average of the key parameters necessary for interpreting the underlying adaptive dynamics.

(d) Utility of unphased data

So far, we have explored whether haplotype alignments contain sufficient information to detect balancing selection, discriminate among its modes and estimate underlying equilibrium frequencies, while also evaluating the advantages of jointly

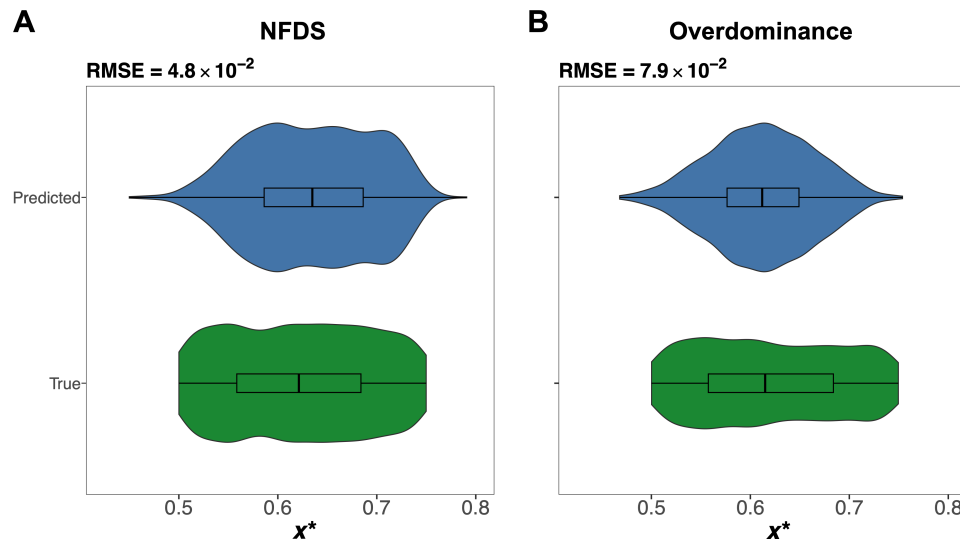


Figure 3. Summaries of distributions for true and predicted equilibrium frequency x^* for (A) NFDS and (B) OD from combined modern and ancient genome variation. For each setting and adaptive parameter, violin plots with embedded box plots together with the root mean square error (RMSE) are displayed.

incorporating ancient and modern variation. However, these inference tasks were tested under idealistic data conditions, assuming that haplotypes can be phased with certainty. Such a scenario is often unattainable for non-model study systems and ancient DNA [66–69]. Furthermore, only a subset of balancing selection modes may contribute to the present distribution of variation in model species [38,70–72], with potentially the most intriguing population dynamics arising from balancing selection in non-model species [42,73,74]. Additionally, in our generated images where haplotypes are sorted, consecutive haplotypes do not necessarily derive from the same individual. As a result, our images of haplotype alignments do not capture genotype states. Further capacity to distinguish between NFDS and OD may be gained by using genotype information, which does not rely on phasing. We therefore examined the efficacy of using a transfer learning-based tool to detect and discriminate among modes of balancing selection from multilocus genotype (MLG) data. In MLG data each SNP is encoded by the number of minor alleles in a diploid genotype, with zero for a homozygous major allele, one for a heterozygous and two for a homozygous minor allele. This encoding differs from haplotype alignments where the allelic state is coded as either zero (major allele) or one (minor allele; see electronic supplementary material, methods). The images generated from MLGs for *TrIdent* are constructed assuming unphased genotype states, which avoids the need for *TrIdent* to rely on phased haplotype variation.

As expected from previous studies [64,75], the classification accuracy when using modern genomic variation declines from 70.17% with phased haplotypes (figure 2A) to 67.80% with MLGs (electronic supplementary material, figure S10A), a reduction largely driven by a decrease in the neutral detection rate from 76.4% to 68.5%. In general, the neutral detection rate for MLGs is lower than that for phased haplotype data across all three input data settings (figure 2 and electronic supplementary material, figure S10). However, the overall accuracy improves slightly when using ancient or combined modern and ancient data with MLGs (compare figure 2B and electronic supplementary material, figure S10B; figures S2C and S10C). This improved accuracy is primarily owing to mode-specific changes in true positive rates. For example, when comparing predictions from MLG with haplotype variation for the combined (ancient and modern) dataset, we observe a 4.7% increase in OD true positives while NFDS true positives increase by 2.3% (figure 2 and electronic supplementary material, figure S10). However, when comparing only ancient datasets, we see a 6.9% increase in true positives for NFDS, while OD true positives increase only by 1.9%. Moreover, as with phased haplotype data, jointly analysing modern and ancient variation using MLGs results in more balanced classification power (electronic supplementary material, figure S11), although OD continues to exhibit significantly lower power to be distinguished from neutrality compared with NFDS (electronic supplementary material, figure S12). These results suggest that distinguishing balancing selection from neutrality is hindered by the increased class overlap when using unphased data, whereas the explicit genotype state information preserved in MLGs enhances the ability to discriminate between modes of balancing selection.

3. Discussion

We have shown that the resource-efficient approach of deep transfer learning provides a promising avenue for improved detection of NFDS, while also permitting characterization of the parameters underlying its adaptive dynamics. Ancient genomic variation can further enhance these inference tasks, and the use of MLGs instead of haplotypes may offer additional capacity to disentangle conflicting signals of different modes of balancing selection. The fact that MLGs show promise is encouraging because even the genotypic values from non-model organisms and ancient samples can be highly uncertain. Such data could be accommodated easily by estimating expected allele counts when constructing MLG data matrices [65]. Our results are also consistent with a recent preliminary analysis, which showed that incorporating ancient variation alongside modern data improves neutral detection rates and yields more balanced classification performance when distinguishing between balancing selection intensities [48]. These findings elucidate the potential for improving genomic data-to-image transformation, model refinement and integration of heterogeneous data sources to enhance evolutionary inference.

One such opportunity is to represent genomic variation more efficiently for downstream modelling. Specifically, the way that *Trident* processes haplotype or MLG alignments for input to the convolutional neural network-based feature extractor could be optimized. Currently, for each haplotype or MLG, the number of minor alleles is counted within a fixed-size sliding window across the genome [64]. Because pre-trained vision models are typically trained on RGB images, *Trident* duplicates grayscale representations of these counts across three channels [64]. This duplication, however, misses the opportunity to encode distinct information in each channel. For MLGs in diploid organisms, where three genotypes are possible—homozygous major, heterozygous and homozygous minor—each channel could instead represent the minor allele counts associated with one genotype state. This approach would decouple genotype contributions to overall pixel values, potentially improving classification performance.

Another refinement can be made in better temporal modelling of input features. When integrating collections of features extracted as vectors from images of ancient variation, we simply concatenated all feature vectors into a single input vector prior to training the classification and regression models. However, these extracted features covary through time across sampled epochs, and explicitly modelling this temporal autocovariation would probably lead to improved inferences [65]. Natural deep learning approaches for capturing such feature dependencies include convolutional and recurrent neural networks [76,77]. Alternatively, after transfer learning-based feature extraction, temporal structure can be incorporated using trend filtering [78–82] or by generating new features via spectral decomposition [52,75,83,84].

As we have shown, incorporating temporal patterns of genomic variation improves our ability to distinguish balancing selection from neutrality. Although we have considered only two mechanisms of balancing selection here, including data from both ancient and modern samples could also improve the detection of other balancing selection modes. Notably, the Stochastic Additive Scale-Concave Fitness Function (SAS-CFF) model [43,44] describes stabilizing selection in abruptly fluctuating environments that can produce dynamics similar to those of NFDS, with alleles experiencing substantial fluctuations in frequency and occasionally drifting close to fixation or loss. By contrast, OD is expected to maintain high variability by preserving balanced polymorphisms in heterozygotes, resulting in less variable allele frequency trajectories. Temporal sampling at sufficient density could capture this difference in variability, allowing more oscillatory modes such as NFDS and SAS-CFF to be distinguished from more stable modes such as OD. Moreover, NFDS actively pulls allele frequencies toward an equilibrium and therefore depends on the current allele frequency, whereas SAS-CFF responds more abruptly to environmental changes and may not correlate as strongly with the current frequency. Explicitly modelling changes in haplotype variation as a function of focal allele frequency dynamics from temporal data could therefore enable multiclass extensions that improve discrimination not only between balancing selection and neutrality, but also among the various modes of balancing selection.

A central goal of this study was to demonstrate the potential of resource-efficient, pre-trained deep transfer learning to detect and distinguish modes of balancing selection while estimating underlying adaptive parameters—all using a relatively small training dataset. However, if resources permit larger-scale data generation and more intensive model training, performance could be improved by fine-tuning the pre-trained model weights for the specific classification or regression tasks described here. Deep learning architectures are inherently data-hungry and may require substantially more training data to realise these gains. In the context of detecting positive selection, for example, the *Trident* framework has shown that fine-tuning can improve classification accuracy by over 5% [64], and similar improvements are likely for regression tasks.

Deep learning models also excel at integrating heterogeneous data sources, an approach often referred to as multimodal artificial intelligence [59], offering new opportunities to enhance detection of NFDS. One promising approach involves incorporating information about deleteriousness or allelic pathogenicity. Balancing selection can interfere with negative selection, allowing deleterious variants to accumulate near selected loci—a phenomenon termed ‘sheltered genetic load’ [85–88]. Empirical evidence for this process has been found in humans [89] and plants [90,91]. The magnitude of deleterious load could be quantified using nonsynonymous variation or pathogenicity scores [92]. These estimates can be generated alongside genetic data for model training by using forward-time simulators that incorporate codon-level mutations and associated distributions of fitness effects [51].

This is a transformative moment for the field of evolutionary genomics. With the ability to simulate genetic data under virtually any evolutionary scenario and the increasing power of deep learning to extract meaningful patterns, we are now equipped to tackle questions that were once out of reach. The synergy between simulation-based data generation, heterogeneous genomic inputs and advanced modelling unlocks new frontiers for evolutionary inference. This work contributes to that momentum by outlining a framework to enhance detection of adaptive processes, particularly balancing selection, and by pointing to scalable strategies for decoding its mechanism across diverse systems.

4. Methods

(a) Data generation

We used the forward-time simulator SLiM v4.3 [51] to generate synthetic data for training a model to discriminate among modes of balancing selection and neutrality, as well as additional data to test model performance. To simulate neutral scenarios, we modelled a human-inspired demographic history involving African, European and East Asian populations, all interconnected through migration following population splits (electronic supplementary material, figure S13) [93,94]. We evolved sequences of length 1.1 megabases [95], using a mutation rate of 2.36×10^{-8} per base per generation and a recombination rate of 10^{-8} , for 5919 generations following a burn-in period of $10 N_e = 73\,105$ generations to seed variation across the modelled sequences [51,93]. Using the ‘recapitation’ method in pyslim [96], we prepended a coalescence simulation of the burn-in period to ensure there was coalescence [51].

We extracted the sequence of genealogies across the simulated region using tree-sequence recording. We sampled 216 modern haplotypes (corresponding to 108 diploid individuals) from the present-day African population, matching the sample size of this population in the 1000 Genomes Project Phase 3 dataset [97]. Additionally, using *tskit* [98], we sampled 20 ancient haplotypes (10 diploid individuals) at each of eight time points, uniformly spaced every 50 generations over the 400 generations preceding the present. This sampling scheme for ancient individuals reflects sample sizes that are currently attainable for some organisms [99,100].

To simulate balancing selection scenarios, we retained all the genetic, demographic and sampling parameters from the neutral simulations but introduced a selected allele with per-generation selection coefficient s at the 550-kilobase position (the centre of the sequence) on a single haplotype 1000 generations prior to sampling.

Under NFDS, the relative fitness of the mutant allele was set to $1 + x^* - p$, where x^* is the equilibrium frequency of the mutant allele and p is its current frequency. The relative fitness of the ancestral allele was set to one. Under OD, the relative fitness values of the homozygous ancestral, heterozygous and homozygous mutant genotypes were set to 1, $1 + hs$ and $1 + s$, respectively, where $h = x^* / (2x^* - 1)$ is the dominance parameter and x^* is again the mutant equilibrium frequency [31,101].

For both balancing selection scenarios, values of $\log_{10}(s) \in [-3, -1]$ and $x^* \in (0.5, 0.75]$ were drawn uniformly at random. This simulation setting is relatively challenging owing to substantial class overlap: with small selection coefficients, balancing selection may appear similar to neutrality in the genome, and within the same range of equilibrium frequencies, distinguishing between the two modes of balancing selection may lack sufficient supporting signal (electronic supplementary material, figure S2).

(b) Classifier construction, training and assessment

We retooled the *TrIdent*[*InceptionResNetV2* (IRV2)] model [54], which was originally designed for the binary problem of distinguishing between selective sweeps from positive selection and neutrality, to address the three-class problem of discriminating among neutrality, OD and NFDS. *TrIdent* operates on image representations of haplotype alignments and extracts features from these images using transfer learning.

Briefly, transfer learning here involves applying a model pre-trained on a large, unrelated dataset to generate lower-dimensional feature representations that capture a broad range of characteristics useful for distinguishing among heterogeneous image classes. This approach enables the rapid generation of a robust set of features that has proven effective in complex categorization settings, typically offering resilience in tasks with few samples, better generalization performance and significant energy savings [102–104].

For feature extraction, we selected the IRV2 architecture [105,106], a deep learning model pre-trained on large image datasets. IRV2 combines residual connections, which improve information flow during training, with inception modules, which enable multi-scale feature identification, making it well-suited for complex image classification tasks.

We simulated 1000 training, validation and test replicates for each class, resulting in a total of 9000 replicates, which is one or two orders of magnitude less training data than typically employed and required to adequately train such deep learning for detecting adaptive events [41,46,52,107–110]. Each replicate was processed through *TrIdent* to obtain a vector of 1536 features from the pre-trained IRV2 architecture that best represent the data through transfer learning. To discriminate among neutrality, OD and NFDS, we trained a logistic regression classifier on the 1000 training observations per class, incorporating an elastic net penalty to control model complexity and sparsity [111].

To select the optimal elastic net hyperparameters, we evaluated each model on the 1000 validation observations per class. The L_1 - and L_2 -norm hyperparameters were chosen as $\alpha \in \{0.0, 0.1, \dots, 1.0\}$ and $\lambda \in \{10^{-6}, 10^{-5}, \dots, 10^5\}$, respectively. The optimal hyperparameter combination was selected to minimize the categorical cross-entropy loss on the validation set [99].

Using the optimal model, we evaluated the effectiveness of this deep transfer learning approach in distinguishing among the three evolutionary scenarios, based on 1000 test observations per class. We considered three model variants: one using features from only modern samples, one using features from only ancient samples, and one using features from both modern and ancient samples. Moreover, for each model variant, we also considered using images based on haplotypes or those based on MLGs. Each sampling epoch—eight ancient and one modern—yielded 1536 extracted features via transfer learning. In the ancient-only setting, a total of $8 \times 1536 = 12\,288$ features was obtained by stacking the eight feature vectors. In the combined modern and ancient setting, a total of $9 \times 1536 = 13\,824$ features was obtained by stacking the eight ancient feature vectors with the modern feature vector. These two scenarios, owing to their high-dimensional input space, benefit significantly from elastic net regularization. The optimal hyperparameters for each trained classifier are reported in electronic supplementary material, table S1.

(c) Regressor construction, training and assessment

As both NFDS and OD are functions of x^* , we trained a model that estimates the equilibrium frequency x^* regardless of the true balancing selection mode. To develop a regression model for predicting x^* , we followed the regression model architecture that had previously yielded optimal prediction accuracy for *TrIdent* [64]. Because the adaptive parameter prediction task within *TrIdent* requires a nonlinear model after transfer learning-based feature extraction, we adopted the previously identified optimal feed-forward neural network architecture used in the *TrIdent* regressor [64]. These architectures include an input layer with 1536 nodes (corresponding to the number of extracted features), four hidden layers and an output layer with a single node and linear activation. For x^* , we implemented the following: the first three hidden layers have 500 nodes with a nonlinear tanh activation and the fourth hidden layer has 200 nodes also with a nonlinear tanh activation.

We then pooled the 1000 training observations from each of the balancing selection classes to create a training set of 2000 observations and also combined the 1000 validation observations from each of these classes to construct a validation set of 2000 observations. The model was implemented in TensorFlow [112] and trained by minimizing the mean squared error using the Adam optimizer [113] with a default learning rate of 10^{-3} for 50 training epochs and a batch size of 50. Elastic net regularization was applied to the weights of each layer, and the optimal combination of L_1 - and L_2 -penalties was selected based on the pair that minimized mean squared error on the validation set [114], using the same hyperparameter grid as for the classifier. The optimal hyperparameters for each trained regressor are reported in electronic supplementary material, table S1.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. All data associated with this manuscript have been deposited on Dryad and can be accessed at the following link: [115].

Supplementary material is available online [116].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. C.G.S.: formal analysis, investigation, methodology, validation, visualization, writing—review and editing; A.L.C.d.S.: formal analysis, investigation, methodology, validation, visualization, writing—review and editing; S.P.A.: methodology, software, writing—review and editing; M.F.: conceptualization, funding acquisition, project administration, supervision, writing—review and editing; M.DeG.: conceptualization, funding acquisition, project administration, supervision, writing—original draft, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein

Conflict of interest declaration. We declare we have no competing interests.

Funding. This work was supported by NIH grant R35GM125890, by NSF grants DEB-1949268 and DBI-2130666, and by UKRI NERC grant NE/Y003519/1. Computations for this research were performed using services provided by Research Computing at Florida Atlantic University.

Acknowledgements. We thank three anonymous referees for their valuable comments that improved this work.

References

- Cheng X, Steinrücken M. 2024 Population genomic scans for natural selection and demography. *Annu. Rev. Genet.* **58**, 319–339. (doi:10.1146/annurev-genet-111523-102651)
- Ruzicka F *et al.* 2025 A century of theories of balancing selection. *bioRxiv* 2025.02.12.637871. (doi:10.1101/2025.02.12.637871)
- Gloss AD, Whiteman NK. 2016 Balancing selection: walking a tightrope. *Curr. Biol.* **26**, R73–R76. (doi:10.1016/j.cub.2015.11.023)
- Hudson RR, Kreitman M, Aguadé M. 1987 A test of neutral molecular evolution based on nucleotide data. *Genetics* **116**, 153–159. (doi:10.1093/genetics/116.1.153)
- Tajima F. 1989 Statistical method for testing the neutral mutation hypothesis by DNA polymorphism. *Genetics* **123**, 585–595. (doi:10.1093/genetics/123.3.585)
- Fu YX, Li WH. 1993 Statistical tests of neutrality of mutations. *Genetics* **133**, 693–709. (doi:10.1093/genetics/133.3.693)
- Simonsen KL, Churchill GA, Aquadro CF. 1995 Properties of statistical tests of neutrality for DNA polymorphism data. *Genetics* **141**, 413–429. (doi:10.1093/genetics/141.1.413)
- Fay JC, Wu CI. 2000 Hitchhiking under positive Darwinian selection. *Genetics* **155**, 1405–1413. (doi:10.1093/genetics/155.3.1405)
- Sabeti PC *et al.* 2002 Detecting recent positive selection in the human genome from haplotype structure. *Nature* **419**, 832–837. (doi:10.1038/nature01140)
- Innan H. 2006 Modified Hudson–Kreitman–Aguadé test and two-dimensional evaluation of neutrality tests. *Genetics* **173**, 1725–1733. (doi:10.1534/genetics.106.056242)
- Voight BF, Kudaravalli S, Wen X, Pritchard JK. 2006 A map of recent positive selection in the human genome. *PLoS Biol.* **4**, e72. (doi:10.1371/journal.pbio.0040072)
- Ferrer-Admetlla A, Liang M, Korneliussen T, Nielsen R. 2014 On detecting incomplete soft or hard selective sweeps using haplotype structure. *Mol. Biol. Evol.* **31**, 1059–1065. (doi:10.1093/molbev/msu077)
- Garud NR, Messer PW, Buzbas EO, Petrov DA. 2015 Recent selective sweeps in North American *Drosophila melanogaster* show signatures of soft sweeps. *PLoS Genet.* **11**, e1005004. (doi:10.1371/journal.pgen.1005004)
- Torres R, Szpiech ZA, Hernandez RD. 2018 Human demographic history has amplified the effects of background selection across the genome. *PLoS Genet.* **4**, e1007387. (doi:10.1371/journal.pgen.1007387)
- Szpiech ZA, Novak TE, Bailey NP, Stevison LS. 2021 Application of a novel haplotype-based scan for local adaptation to study high-altitude adaptation in rhesus macaques. *Evol. Lett.* **4**, 408–421. (doi:10.1002/evl3.232)
- Slatkin M. 2008 Linkage disequilibrium — understanding the evolutionary past and mapping the medical future. *Nat. Rev. Genet.* **9**, 477–485. (doi:10.1038/nrg2361)
- Price AL *et al.* 2008 Long-range LD can confound genome scans in admixed populations. *Am. J. Hum. Genet.* **83**, 132–135. (doi:10.1016/j.ajhg.2008.06.005)
- Cavedon M *et al.* 2019 Genomics, environment and balancing selection in behaviourally bimodal populations: the caribou case. *Mol. Ecol.* **28**, 1946–1963. (doi:10.1111/mec.15039)
- Christie MR, McNickle GG. 2023 Negative frequency dependent selection unites ecology and evolution. *Ecol. Evol.* **13**, e10327. (doi:10.1002/ece3.10327)
- Kim Y, Stephan W. 2002 Detecting a local signature of genetic hitchhiking along a recombining chromosome. *Genetics* **160**, 757–777. (doi:10.1093/genetics/160.2.765)
- Nielsen R, Williamson S, Kim Y, Hubisz MJ, Clark AG, Bustamante C. 2005 Genomic scans for selective sweeps using SNP data. *Genome Res.* **15**, 1566–1575. (doi:10.1101/gr.4252305)
- Chen H, Patterson N, Reich D. 2010 Population differentiation as a test for selective sweeps. *Genome Res.* **20**, 393–402. (doi:10.1101/gr.100545.109)
- Huber CD, DeGiorgio M, Hellmann I, Nielsen R. 2015 Detecting recent selective sweeps while controlling for mutation rate and background selection. *Mol. Ecol.* **25**, 142–156. (doi:10.1111/mec.13351)
- Vy HMT, Kim Y. 2015 A composite-likelihood method for detecting incomplete selective sweep from population genomic data. *Genetics* **200**, 633–649. (doi:10.1534/genetics.115.175380)
- Racimo F. 2016 Testing for ancient selection using cross-population allele frequency differentiation. *Genetics* **202**, 733–750. (doi:10.1534/genetics.115.178095)
- Lee KM, Coop G. 2017 Distinguishing among modes of convergent adaptation using population genomic data. *Genetics* **207**, 1591–1619. (doi:10.1534/genetics.117.300417)
- Setter D, Mousset S, Cheng X, Nielsen R, DeGiorgio M, Hermisson J. 2020 VolcanoFinder: Genomic scans for adaptive introgression. *PLoS Genet.* **15**, e1008867. (doi:10.1371/journal.pgen.1008867)
- DeGiorgio M, Szpiech ZA. 2022 A spatially aware likelihood test to detect sweeps from haplotype distributions. *PLoS Genet.* **18**, e1010134. (doi:10.1371/journal.pgen.1010134)
- Beaumont MA, Balding DJ. 2004 Identifying adaptive genetic divergence among populations from genome scans. *Mol. Ecol.* **13**, 969–980. (doi:10.1111/j.1365-294x.2004.02125.x)
- Sugden LA, Atkinson EG, Fischer AP, Rong S, Henn BM, Ramachandran S. 2017 Localization of adaptive variants in human genomes using averaged one-dependence estimation. *Nat. Commun.* **9**, 703. (doi:10.1038/s41467-018-03100-7)

31. DeGiorgio M, Lohmueller KE, Nielsen R. 2014 A model-based approach for identifying signatures of ancient balancing selection in genetic data. *PLoS Genet.* **10**, e1004561. (doi:10.1371/journal.pgen.1004561)
32. Siewert KM, Voight BF. 2017 Detecting long-term balancing selection using allele frequency correlation. *Mol. Biol. Evol.* **34**, 2996–3005. (doi:10.1093/molbev/msx209)
33. Siewert KM, Voight BF. 2020 BetaScan2: standardized statistics to detect balancing selection utilizing substitution data. *Genome Biol. Evol.* **12**, 3873–3877. (doi:10.1093/gbe/evaa013)
34. Bitarello BD, de Filippo C, Teixeira JC, Schmidt JM, Kleinert P, Meyer D, Andrés AM. 2018 Signatures of long-term balancing selection in human genomes. *Genome Biol. Evol.* **10**, 939–955. (doi:10.1093/gbe/evy054)
35. Cheng X, DeGiorgio M. 2019 Detection of shared balancing selection in the absence of trans-species polymorphism. *Mol. Biol. Evol.* **36**, 177–199. (doi:10.1093/molbev/msy202)
36. Cheng X, DeGiorgio M. 2020 Flexible mixture model approaches that accommodate footprint size variability for robust detection of balancing selection. *Mol. Biol. Evol.* **37**, 3267–3291. (doi:10.1093/molbev/msaa134)
37. Cheng X, DeGiorgio M. 2022 BalLeRMix +: mixture model approaches for robust joint identification of both positive selection and long-term balancing selection. *Bioinformatics* **38**, 861–863. (doi:10.1093/bioinformatics/btab720)
38. Bitarello BD, Brandt DYC, Meyer D, Andrés AM. 2023 Inferring balancing selection from genome-scale data. *Genome Biol. Evol.* **15**, evad032. (doi:10.1093/gbe/evad032)
39. Gao Z, Przeworski M, Sella G. 2015 Footprints of ancient-balanced polymorphisms in genetic variation data from closely related species. *Evolution* **69**, 431–436. (doi:10.1111/evo.12567)
40. Soni V, Jensen JD. 2024 Temporal challenges in detecting balancing selection from population genomic data. *G3* **14**, jkae069. (doi:10.1093/g3journal/jkae069)
41. Isildak U, Stella A, Fumagalli M. 2021 Distinguishing between recent balancing selection and incomplete sweep using deep neural networks. *Mol. Ecol. Resour.* **21**, 2706–2718. (doi:10.1111/1755-0998.13379)
42. Schield DR *et al.* 2022 The roles of balancing selection and recombination in the evolution of rattlesnake venom. *Nat. Ecol. Evol.* **6**, 1367–1380. (doi:10.1038/s41559-022-01829-5)
43. Gillespie JH. 1978 A general model to account for enzyme variation in natural populations. V. The SAS-CFF model. *Theor. Popul. Biol.* **14**, 1–45. (doi:10.1016/0040-5809(78)90002-3)
44. Gillespie JH. 1992 *The causes of molecular evolution*. New York, NY: Oxford University Press. (doi:10.1093/oso/9780195068832.001.0001)
45. Lin K, Li H, Schlötterer C, Futschik A. 2011 Distinguishing positive selection from neutral evolution: boosting the performance of summary statistics. *Genetics* **187**, 229–244. (doi:10.1534/genetics.110.122614)
46. Sheehan S, Song YS. 2016 Deep learning for population genetic inference. *PLoS Comput. Biol.* **12**, 1–28. (doi:10.1371/journal.pcbi.1004845)
47. Schrider DR, Kern AD. 2018 Supervised machine learning for population genetics: a new paradigm. *Trends Genet.* **34**, 301–312. (doi:10.1016/j.tig.2017.12.005)
48. Korfmann K, Gaggiotti OE, Fumagalli M. 2023 Deep learning in population genetics. *Genome Biol. Evol.* **15**, evad008. (doi:10.1093/gbe/evad008)
49. Ahlquist KD, Sugden LA, Ramachandran S. 2023 Enabling interpretable machine learning for biological data with reliability scores. *PLoS Comput. Biol.* **19**, e1011175. (doi:10.1371/journal.pcbi.1011175)
50. Baumdicker F *et al.* 2021 Efficient ancestry and mutation simulation with msprime 1.0. *Genetics* **220**, iyab229. (doi:10.1093/genetics/iyab229)
51. Haller BC, Messer PW. 2023 SLiM 4: multispecies eco-evolutionary modeling. *Am. Nat.* **201**, E127–E139. (doi:10.1086/723601)
52. Arnab SP, Amin MR, DeGiorgio M. 2023 Uncovering footprints of natural selection through spectral analysis of genomic summary statistics. *Mol. Biol. Evol.* **40**, msad157. (doi:10.1093/molbev/msad157)
53. Smith CCR, Patterson G, Ralph PL, Kern AD. 2024 Estimation of spatial demographic maps from polymorphism data using a neural network. *Mol. Ecol. Resour.* **24**, e14005. (doi:10.1111/1755-0998.14005)
54. LeCun Y, Bengio Y. 1995 Convolutional networks for images, speech, and time series. In *The handbook of brain theory and neural networks* (ed. MA Arbib), pp. 255–258. Cambridge, MA: MIT.
55. LeCun Y, Bengio Y, Hinton G. 2015 Deep learning. *Nature* **521**, 436–444. (doi:10.1038/nature14539)
56. Xu Y, Xiao T, Zhang J, Yang K, Zhang Z. 2014 Scale-invariant convolutional neural networks. *arXiv* 1411.6369. (doi:10.48550/arXiv.1411.6369)
57. Cybenko G. 1989 Approximation by superpositions of a sigmoidal function. *Math. Control. Signal. Systems* **2**, 303–314. (doi:10.1007/BF02551274)
58. Hornik K, Stinchcombe M, White H. 1989 Multilayer feedforward networks are universal approximators. *Neural Netw.* **2**, 359–366. (doi:10.1016/0893-6080(89)90020-8)
59. Kang M, Ko E, Mersha TB. 2021 A roadmap for multi-omics data integration using deep learning. *Brief. Bioinform.* **23**, bbab454. (doi:10.1093/bib/bbab454)
60. Andrés AM. 2011 Balancing selection in the human genome. In *Encyclopedia of life sciences*. Chichester, UK: John Wiley and Sons. (doi:10.1002/9780470015902.a0022863)
61. Fijarczyk A, Babik W. 2015 Detecting balancing selection in genomes: limits and prospects. *Mol. Ecol.* **24**, 3529–3545. (doi:10.1111/mec.13226)
62. Grealey J, Lannelongue L, Saw WY, Marten J, Meric G, Ruiz-Carmona S, Inouye M. 2022 The carbon footprint of bioinformatics. *Mol. Biol. Evol.* **39**, msac034. (doi:10.1101/2021.03.08.434372)
63. Bozinovski S. 2020 Reminder of the first paper on transfer learning in neural networks, 1976. *Informatica* **44**, 3. (doi:10.31449/inf.v44i3.2828)
64. Arnab SP, Campelo Dos Santos AL, Fumagalli M, DeGiorgio M. 2025 Efficient detection and characterization of targets of natural selection using transfer learning. *Mol. Biol. Evol.* **42**, msaf094. (doi:10.1093/molbev/msaf094)
65. Whitehouse LS, Schrider DR. 2023 Timesweeper: accurately identifying selective sweeps using population genomic time series. *Genetics* **224**, iyad084. (doi:10.1093/genetics/iyad084)
66. Ros-Freixedes R, Battagin M, Johnsson M, Gorjanc G, Mileham AJ, Rounsley SD, Hickey JM. 2018 Impact of index hopping and bias towards the reference allele on accuracy of genotype calls from low-coverage sequencing. *Genet. Sel. Evol.* **50**, 64. (doi:10.1186/s12711-018-0436-4)
67. Günther T, Nettelblad C. 2019 The presence and impact of reference bias on population genomic studies of prehistoric human populations. *PLoS Genet.* **15**, e1008302. (doi:10.1101/487983)
68. Dehasque M *et al.* 2020 Inference of natural selection from ancient DNA. *Evol. Lett.* **4**, 94–108. (doi:10.1002/evl3.165)
69. Tretmanis JM, Jay F, Avila-Árcos MC, Huerta-Sanchez E. 2023 Simulation-based benchmarking of ancient haplotype inference for detecting population structure. *bioRxiv* 2023.09.28.560049. (doi:10.1101/2023.09.28.560049)
70. Bergland AO, Behrman EL, O'Brien KR, Schmidt PS, Petrov DA. 2014 Genomic evidence of rapid and stable adaptive oscillations over seasonal time scales in *Drosophila*. *PLoS Genet.* **10**, e1004775. (doi:10.1371/journal.pgen.1004775)
71. Chakraborty M, Fry JD. 2016 Evidence that environmental heterogeneity maintains a detoxifying enzyme polymorphism in *Drosophila melanogaster*. *Curr. Biol.* **26**, 219–223. (doi:10.1016/j.cub.2015.11.049)

72. Liebman KA, Pinto J, Valle J, Palomino M, Vizcaino L, Brogdon W, Lenhart A. 2015 Novel mutations on the *ace-1* gene of the malaria vector *Anopheles albimanus* provide evidence for balancing selection in an area of high insecticide resistance in Peru. *Malar. J.* **14**, 74. (doi:10.1186/s12936-015-0599-1)
73. Archie EA, Henry T, Maldonado JE, Moss CJ, Poole JH, Pearson VR, Murray S, Alberts SC, Fleischer RC. 2010 Major histocompatibility complex variation and evolution at a single, expressed DQA locus in two genera of elephants. *Immunogenetics* **62**, 85–100. (doi:10.1007/s00251-009-0413-8)
74. Lamichaney S *et al.* 2016 Structural genomic changes underlie alternative reproductive strategies in the ruff (*Philomachus pugnax*). *Nat. Genet.* **48**, 84–88. (doi:10.1038/ng.3430)
75. Amin MR, Hasan M, DeGiorgio M. 2024 Digital image processing to detect adaptive evolution. *Mol. Biol. Evol.* **41**, msae242. (doi:10.1093/molbev/msae242)
76. Williams RJ, Hinton GE, Rumelhardt DE. 1986 Learning representations by back-propagating errors. *Nature* **323**, 533–536. (doi:10.1038/323533a0)
77. Lecun Y, Bottou L, Bengio Y, Haffner P. 1998 Gradient-based learning applied to document recognition. *Proc. IEEE* **86**, 2278–2324. (doi:10.1109/5.726791)
78. Kim SJ, Koh K, Boyd S, Gorinevsky D. 2009 l_1 trend filtering. *SIAM. Rev.* **2**, 339–360. (doi:10.1137/070690274)
79. Hawkins DM, Maboudou-Tchao EM. 2013 Smoothed linear modeling for smooth spectral data. *Int. J. Spectrosc.* **2013**, 1–8. (doi:10.1155/2013/604548)
80. Tibshirani RJ. 2014 Adaptive piecewise polynomial estimation via trend filtering. *Ann. Statist.* **42**, 285–323. (doi:10.1214/13-AOS1189)
81. Wang YX, Sharpnack J, Smola AJ, Tibshirani RJ. 2016 Trend filtering on graphs. *J. Mach. Learn. Res.* **17**, 1–41. (doi:10.5555/2946645.3007058)
82. Mughal MR, DeGiorgio M. 2019 Localizing and classifying adaptive targets with trend filtered regression. *Mol. Biol. Evol.* **36**, 252–270. (doi:10.1093/molbev/msy205)
83. Ramsey JO, Silverman BW. 2005 *Functional data analysis*, 2nd edn. New York, NY: Springer.
84. Mughal MR, Koch H, Huang J, Chiaromonte F, DeGiorgio M. 2021 Learning the properties of adaptive regions with functional data analysis. *PLoS. Genet.* **16**, e1008896. (doi:10.1101/834010)
85. Charlesworth B, Nordborg M, Charlesworth D. 1997 The effects of local selection, balanced polymorphism and background selection on equilibrium patterns of genetic diversity in subdivided populations. *Genet. Res.* **70**, 155–174. (doi:10.1017/S0016672397002954)
86. Uyenoyama MK. 1997 Genealogical structure among alleles regulating self-incompatibility in natural populations of flowering plants. *Genetics* **147**, 1389–1400. (doi:10.1093/genetics/147.3.1389)
87. Uyenoyama MK. 2005 Evolution under tight linkage to mating type. *New Phytol.* **165**, 63–70. (doi:10.1111/j.1469-8137.2004.01246.x)
88. Hartfield M, Otto SP. 2011 Recombination and hitchhiking of deleterious alleles. *Evolution* **65**, 2421–2434. (doi:10.1111/j.1558-5646.2011.01311.x)
89. Lenz TL, Spirin V, Jordan DM, Sunyaev SR. 2016 Excess of deleterious mutations around *HLA* genes reveals evolutionary cost of balancing selection. *Mol. Biol. Evol.* **33**, 2555–2564. (doi:10.1093/molbev/msw127)
90. Llaurens V, Gonthier L, Billiard S. 2009 The sheltered genetic load linked to the *S* locus in plants: new insights from theoretical and empirical approaches in sporophytic self-incompatibility. *Genetics* **183**, 1105–1118. (doi:10.1534/genetics.109.102707)
91. Le Veve A *et al.* 2023 Long-term balancing selection and the genetic load linked to the self-incompatibility locus in *Adabidopsis helleri* and *A. lyrata*. *Mol. Biol. Evol.* **40**, msad120. (doi:10.1093/molbev/msad120)
92. Le Veve A, Genete M, Lepers-Blassiau C, Ponitzki C, Poux C, Vekemans X, Ruand E, Castric V. 2024 The genetic architecture of the load linked to dominant and recessive self-incompatibility alleles in *Arabidopsis halleri* and *Arabidopsis lyrata*. *eLife* **13**, RP94972. (doi:10.7554/eLife.94972.3.sa2)
93. Gravel S *et al.* 2011 Demographic history and rare allele sharing among human populations. *Proc. Natl Acad. Sci. USA* **108**, 11983–11988. (doi:10.1073/pnas.1019276108)
94. Gower G *et al.* 2022 Demes: a standard format for demographic models. *Genetics* **222**, iyac131. (doi:10.1093/genetics/iyac131)
95. Schrider DR, Kern AD. 2016 S/HIC: robust identification of soft and hard sweeps using machine learning. *PLoS Genet.* **12**, e1005928. (doi:10.1371/journal.pgen.1005928)
96. Gopalan S, Rodrigues MF, Ralph PL, Haller BC. Bridging forward-in-time and coalescent simulations using pyslim. *bioRxiv*. (doi:10.1101/2025.09.30.679676)
97. The 1000 Genomes Project Consortium. 2015 A global reference for human genetic variation. *Nature* **526**, 68–74. (doi:10.1038/nature15393)
98. Kelleher J, Etheridge AM, McVean G. 2016 Efficient coalescent simulation and genealogical analysis for large sample sizes. *PLoS Comput. Biol.* **12**, e1004842. (doi:10.1371/journal.pcbi.1004842)
99. Brunson K, Reich D. 2019 The promise of paleogenomics beyond our own species. *Trends Genet.* **35**, 319–329. (doi:10.1016/j.tig.2019.02.006)
100. Ávila-Arcos MC, Raghavan M, Schlebusch C. 2023 Going local with ancient DNA: a review of human histories from regional perspectives. *Science* **482**, 53–58. (doi:10.1126/science.adh8140)
101. Gillespie JH. 2004 *Population genetics: a concise guide*. Baltimore, MD: JHU Press.
102. Yosinski J, Clune J, Bengio Y, Lipson H. 2014 How transferable are features in deep neural networks? In *NIPS'14: Proceedings of the 28th International Conference on Neural Information Processing Systems*, Montreal, Canada, 8–13 December 2014, vol. 2, pp. 3320–3328, Cambridge, MA: MIT. (doi:10.48550/arXiv.1411.1792)
103. Azizpour H, Razavian AS, Sullivan J, Maki A, Carlsson S. 2015 Factors of transferability for a generic ConvNet representation. *IEEE Trans. Pattern Anal. Mach. Intell.* **38**, 1790–1802. (doi:10.1109/TPAMI.2015.2500224)
104. Beyer L, Hénaff OJ, Kolesnikov A, Zhai X, van den Oord A. 2020 Are we done with ImageNet? *arXiv* 2006.07159. (doi:10.48550/arXiv.2006.07159)
105. Szegedy C, Ioffe S, Vanhoucke V, Alemi A. 2017 Inception-v4, Inception-ResNet and the Impact of Residual Connections on Learning. vol. **31**. (doi:10.1609/aaai.v31i1.11231)
106. Wang J, He X, Faming S, Lu G, Cong H, Jiang Q. 2021 A real-time bridge crack detection method based on an improved inception-resnet-v2 structure. *IEEE Access* **9**, 93209–93223. (doi:10.1109/access.2021.3093210)
107. Gower G, Picazo PI, Fumagalli M, Racimo F. 2021 Detecting adaptive introgression in human evolution using convolutional neural networks. *eLife* **10**, e64669. (doi:10.7554/eLife.64669)
108. Lauterbur ME, Munch K, Enard D. 2023 Versatile detection of diverse selective sweeps with flex-sweep. *Mol. Biol. Evol.* **40**, msad139. (doi:10.1093/molbev/msad139)
109. Kern AD, Schrider DR. 2018 diploS/HIC: an updated approach to classifying selective sweeps. *G3* **8**, 1959–1970. (doi:10.1534/g3.118.200262)
110. Torada L, Lorenzon L, Beddis A, Isildak U, Pattini L, Mathieson S, Fumagalli M. 2019 ImaGene: a convolutional neural network to quantify natural selection from genomic data. *BMC Bioinform.* **20**, 337. (doi:10.1186/s12859-019-2927-x)
111. Zou H, Hastie T. 2005 Regularization and variable selection via the elastic net. *J. R. Stat. Soc. B* **67**, 301–332. (doi:10.1111/j.1467-9868.2005.00503.x)
112. Abadi M *et al.* 2015 TensorFlow: large-scale machine learning on heterogeneous systems. *Software available from tensorflow.org*. See <https://www.tensorflow.org/>.
113. Kingma D, Ba J. 2015 Adam: a method for stochastic optimization. In *3rd Int. Conf. of Learning Representations (ICLR)*, San Diego, CA, 7–9 May 2015. (doi:10.48550/arXiv.1412.6980)
114. Hastie T, Tibshirani R, Friedman J. 2009 *The elements of statistical learning: data mining, inference, and prediction*, 2nd edn. New York, NY: Springer.
115. Santander CG, Campelo dos Santos AL, Arnab SP, Fumagalli M, DeGiorgio M. 2025 Negative frequency-dependent selection: a positive outlook with deep learning [Dataset]. *Dryad Digital Repository* (doi:10.5061/dryad.w3r22813q)
116. Santander C, Campelo dos Santos AL, Arnab SP, Fumagalli M, DeGiorgio M. 2026 Supplementary material from: Negative frequency-dependent selection: a positive outlook with deep learning. Figshare. (doi:10.6084/m9.figshare.c.8447772)